

Related Work and Data Description

Capstone Project (CAP182) – SS2 Part A

Related Literature 1

Title	Predicting Customer Churn on OTT Platforms: Customers with Subscription of Multiple Service Providers
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Harvard reference	Mohan, M. and Jadhav, A. (2022) 'Predicting Customer Churn on OTT Platforms: Customers with Subscription of Multiple Service Providers', <i>Journal of Information and Organizational Sciences</i> , 46(2), pp.433–451. doi: 10.31341/jios.46.2.10.
Link	https://doi.org/10.31341/jios.46.2.10
Title	Explainable AI for Customer Churn Prediction in the Energy Sector Using Ensemble Machine Learning Models
Harvard reference	Al Mamun, A., Tanoor, M.F.S., Masum, A.K.M., Bhuiyan, T. and Hassan, M.M. (2026) 'Explainable AI for Customer Churn Prediction in the Energy Sector Using Ensemble Machine Learning Models', <i>EAI Endorsed Transactions on AI and Robotics</i> , 5. doi: 10.4108/airo.9813.
Link	https://doi.org/10.4108/airo.9813
Modelling	Three ensemble tree-based models were compared: Random Forest (300 trees, max depth 10, Gini criterion), XGBoost (learning rate 0.1, max depth 6, 200 estimators, subsample ratio 0.8), and LightGBM (learning rate 0.05, 250 estimators, feature fraction 0.8). Hyperparameters were tuned via grid search within 5-fold stratified cross-validation. XGBoost was selected as the best-performing model and further explained using SHAP (global) and LIME (local) interpretability techniques.
Preprocessing techniques	Missing numeric values imputed with the mean, missing categorical values imputed with the mode. Categorical features were label encoded. Outliers were capped using the IQR method (values outside $Q1 - 1.5 \times IQR$ to $Q3 + 1.5 \times IQR$). Class imbalance (~9:1 non-churn to churn) was corrected using SMOTE, applied only within each training fold to avoid data leakage.
Datasets used	Kaggle's PowerCo dataset – the same dataset used in this project – combining client data and pricing data via a shared ID column (12,241 combined instances, 9.1% churners).

Key takeaways	(1) This paper uses the exact same PowerCo dataset as this project, so its preprocessing pipeline (imputation, encoding, IQR outlier capping) can be reused as a direct starting point. (2) SMOTE is shown to be an effective, well-documented way to handle this dataset's known class imbalance rather than leaving a model biased toward the majority class. (3) Ensemble tree models (Random Forest, XGBoost, LightGBM) all performed strongly (94% accuracy) on this exact dataset, supporting their use for Part B modelling.
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Related Literature 2

Modelling	Hierarchical Logistic Regression was first used to test the impact of two engineered factors, 'Multiple Subscription' and 'Switching Frequency', on churn. Final churn prediction was then performed and compared across four classifiers: Decision Tree, Random Forest, AdaBoost, and Gradient Boosting. Random Forest produced the best prediction results.
Preprocessing techniques	Feature ranking and selection was performed using Recursive Feature Elimination (RFE) combined with Linear Regression and Ridge Regression ranking methods, to identify the most influential churn factors from the 19 survey items collected.
Datasets used	A primary dataset collected by the authors via a structured questionnaire (19 items covering demographics, OTT usage, and satisfaction factors) from 317 respondents with subscriptions to multiple OTT platforms. This is not a publicly available dataset.
Key takeaways	(1) This is the closest match to STADIOchoice's actual business problem, since it studies OTT/streaming subscribers directly rather than a different industry. (2) It shows that engineered behavioural features (e.g. switching frequency) can meaningfully improve churn prediction, supporting the creation of similar engineered features (e.g. days since last login, search behaviour) once STADIOchoice's own data is available. (3) Random Forest again outperformed simpler models, reinforcing ensemble tree methods as a strong default choice for churn classification.

Related Literature 3

Title	Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models
Harvard reference	Chang, V., Hall, K., Xu, Q.A., Amao, F.O., Ganatra, M.A. and Benson, V. (2024) 'Prediction of Customer Churn Behavior in the Telecommunication Industry Using Machine Learning Models', <i>Algorithms</i> , 17(6), 231. doi: 10.3390/a17060231.
Link	https://doi.org/10.3390/a17060231
Modelling	Five classifiers were trained and compared: Logistic Regression, K-Nearest Neighbour (KNN), Naïve Bayes, Decision Tree, and Random Forest. Random Forest performed best (86.94% accuracy, AUC 0.95). LIME (local explanations) and SHAP (global feature importance) were applied to the best model to interpret which features drove churn predictions.
Preprocessing techniques	Data quality checks were performed for missing values across all variables. A 75%/25% train-test split was used, informed by the dataset's non-churn-to-churn ratio. No rows were removed after the data quality and variable-selection review.
Datasets used	The Telecom Customer Churn dataset from the Maven Analytics platform – 7,043 customer rows and 38 attributes (publicly available; not the dataset used in this project).
Key takeaways	(1) Even with a large, feature-rich telecom dataset, a comparatively simple Random Forest outperformed more complex approaches, suggesting greater model complexity is not automatically better for churn problems. (2) Combining LIME and SHAP gave consistent, complementary explanations – a useful pattern for making STADIOchoice's eventual model results explainable to non-technical stakeholders. (3) The authors identify a lack of demographic data as a real limitation of their model, directly reinforcing this project's decision to explicitly request demographic data from STADIOchoice.

Publicly Available Dataset

Name	PowerCo SME Customer Churn Dataset
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Harvard reference	sumit08 (2024) <i>PowerCo Company</i> , Kaggle. Available at: https://www.kaggle.com/datasets/sumit08/powerco-company
Link	https://www.kaggle.com/datasets/sumit08/powerco-company
Number of instances	14,606 customer records (single table; no separate train/test split provided – a split will be created during modelling in Part B).
Number of feature columns	43 original feature columns, plus 1 target column (churn), for 44 columns in total.
Number of classes (if applicable)	2 classes (binary classification): churned (1) vs. not churned (0). The dataset is imbalanced – 1,419 churned customers (9.72%) vs. 13,187 non-churned customers (90.28%).
Target range (if applicable)	Not applicable – this is a classification problem (churn / no churn), not a regression problem.
Origin	Originally supplied as client and pricing data for a fictional/anonymised European energy retailer ('PowerCo') as part of the BCG (Boston Consulting Group) Data Science Job Simulation, a virtual work-experience programme hosted on the Forage platform. The data was subsequently shared publicly on Kaggle. It is real-structured customer data (consumption, pricing, contract, and churn fields) rather than synthetically generated.